

CALIFORNIA FISH and GAME



"CONSERVATION THROUGH EDUCATION"

VOLUME 82

FALL 1996

NUMBER 4



COMPARATIVE FEEDING HABITS OF JUVENILE CHINOOK SALMON, STEELHEAD, AND SACRAMENTO SQUAWFISH IN THE LOWER AMERICAN RIVER, CALIFORNIA

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We compared diets of juvenile chinook salmon, *Oncorhynchus tshawytscha*, steelhead, *O. mykiss*, and Sacramento squawfish, *Ptychocheilus grandis*, in the lower American River from February through July 1992 and 1993. Chinook salmon and steelhead fed primarily on chironomids (all stages), baetids (adults and nymphs), and hydropsychids (larvae and pupae); their diets overlapped significantly in May and June. Squawfish diet consisted mainly of corixid adults and chironomid larvae and did not overlap significantly with either chinook salmon or steelhead diets. Food habits changed somewhat as the season progressed and the fish increased in size. In 1993, chinook salmon and steelhead fed less than in the previous year and relied more on chironomids. We attribute these differences in diet to the major fluctuations in discharge and lower water temperatures that occurred in the lower American River that year.

INTRODUCTION

The American River, a major tributary of the Sacramento River, is typical of most rivers in the United States in that it has experienced various perturbations such as impoundment, introduction of exotic fish species, and pollution (Allan and Flecker 1993). The only reach of the American River with unobstructed flow is the 39-km section in Sacramento County between Nimbus Dam and the confluence with the Sacramento River. Although flows are controlled by Nimbus and Folsom dams, the lower American River (LAR) still supports a varied fish community, including substantial spawning runs of chinook salmon, *Oncorhynchus tshawytscha*, and steelhead, *O. mykiss*, which are supplemented by fingerlings from Nimbus Fish Hatchery; hatchery-reared chinook fingerlings are trucked downstream to Carquinez Strait for release. The LAR provides critical nursery habitat for these salmonids from hatching in winter and spring to smoltification in late spring into fall.

Water development projects proposed for the LAR, such as extension of the Folsom South Canal and East Bay Municipal Utility District withdrawals, would alter or reduce flows. Understanding the impact of these projects requires thorough knowledge of the ecology of the river's fish assemblage, including trophic relationships.

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As a step in that direction, we conducted an analysis of the diets of juvenile chinook salmon and steelhead in the LAR. We also analyzed food habits of juvenile Sacramento squawfish, *Ptychocheilus grandis*, because it is another abundant predatory species in the LAR whose young frequently have been observed in the same habitats as chinook and steelhead (Brown et al.² 1992). Historically, this species has been considered a competitor with young salmonids (Pintler and Johnson 1958, Burns 1966, Brown and Moyle 1982). Our specific objectives were to compare diets and feeding patterns of juvenile chinook salmon, steelhead, and Sacramento squawfish on a monthly basis for 2 consecutive years during the months that all three species were present in the LAR.

STUDY AREA

The LAR is characterized by alternating bar complex and flatwater habitats, with a gradient of approximately 0.6 m per km. The drainage area consists of 179 km² of mostly urbanized land. Several small streams and storm drains enter the river. At least 28 fish species are found in the LAR (Snider and Titus³ 1994). Among the most abundant species, in addition to chinook salmon, steelhead, and Sacramento squawfish, are the Sacramento sucker, *Catostomus occidentalis*; prickly sculpin, *Cottus asper*; and tule perch, *Hysterocarpus traski*. Other anadromous species found seasonally are the Pacific lamprey, *Lampetra tridentata*; American shad, *Alosa sapidissima*; and striped bass, *Morone saxatilis*.

Mean daily flows during the 1992 study period were relatively constant, approximately 25 m³/s from February through May, then gradually increased to a maximum of 127 m³/s in June (Fig. 1). In contrast, flows during the 1993 study period were higher and more variable; several major fluctuations involved abrupt changes in a short time. Highest flow that year was 430 m³/s in early April followed by an abrupt drop to 30 m³/s, the lowest flow since February.

Water temperatures gradually increased during the study period in both years, but temperatures were considerably higher in 1992, increasing from about 11°C in February to about 19°C in late June (Fig. 2). In 1993, temperatures averaged about 3°C lower than in 1992 throughout the study period.

METHODS

Fish were collected monthly between 1 February and 1 July 1992 and 1993 at three locations: 1) downstream from the H Street bridge (river km 10), 2) Gristmill (river km 17), and 3) above the Sunrise Boulevard bridge (river km 32). These sites

² Brown, L., P. Moyle, and D. Vanicek. 1992. American River studies: Intensive fish surveys, March-June, 1991. Mimeo Report, Department of Wildlife and Fisheries Biology, University of California, Davis, California, USA.

³ Snider, W. and R. Titus. 1994. Fish community survey, lower American River, January-July 1994. Mimeo Report, California Department of Fish and Game, Environmental Services Division, Sacramento, California, USA.

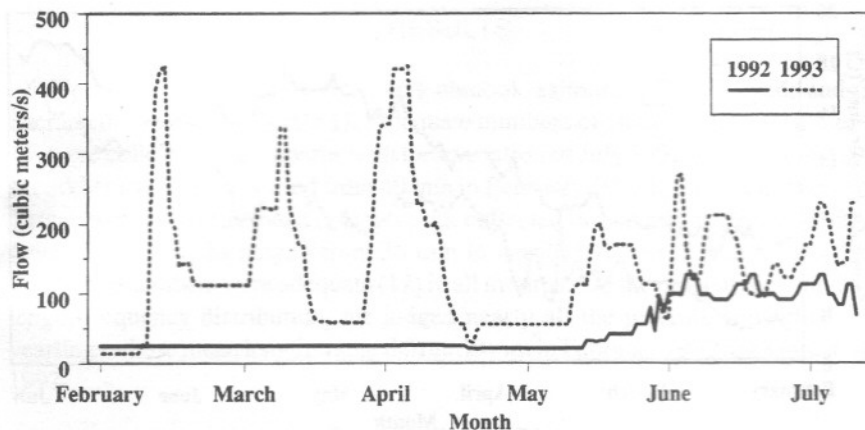


Figure 1. Lower American River mean daily flow from 1 February to 15 July 1992 and 1993. Data provided by California Department of Fish and Game.

were selected because of their accessibility and because they typified the LAR's habitats. We collected fish during the first week of each month between 0900 and 1500 with a 15 x 2-m bag seine with 3-mm square mesh. Our monthly target sample sizes were at least 14 chinook salmon, 14 steelhead, and 12 squawfish, which were based on preliminary stomach analysis (see Hurturbia 1973). Fish were immediately preserved in 10% formalin and transported to the laboratory for analysis. The fork length of each fish was measured to the nearest millimeter and stomach contents were removed, sorted under a dissecting microscope, and weighed to the nearest 0.001 g. For squawfish, which lack a distinct stomach, we analyzed contents in the anterior portion of the alimentary tract from the esophagus to the first bend. Food items were identified to order or family with the aid of Merritt and Cummins (1984), Wang⁴ (1986), and Pennak (1989). Food habits data for each species were pooled on a monthly basis and initially analyzed by frequency of occurrence, numeric, and gravimetric (dry weight) methods (Bowen 1983). To assess the relative importance of food items, we calculated an index of relative importance (IRI) for each food category, as described by Hyslop (1980):

$$IRI = (FN + FW) FO,$$

where

FN = a food item's percentage of the total number of organisms ingested,

FW = a food item's percentage of the total weight of food ingested, and

FO = a food item's percentage frequency of occurrence in all stomachs examined that contained food.

⁴ Wang, J. 1986. Fishes of the Sacramento-San Joaquin Estuary and adjacent waters, California: A guide to the early life histories. Technical Report No. 9, California Department of Water Resources, Sacramento, California, USA.

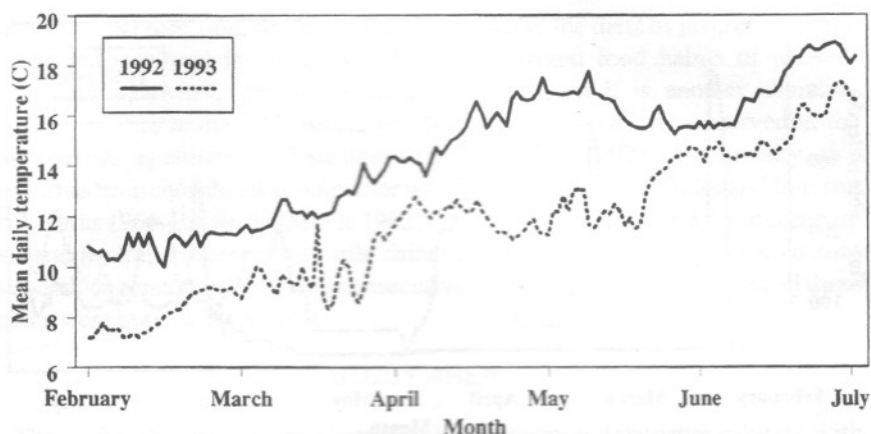


Figure 2. Lower American River mean daily temperature from 15 February to 15 July 1992 and 1993. Data provided by California Department of Fish and Game.

To make dietary comparisons, we converted the IRI values of each food item to percentages based on total IRIs for each month. Food items that occurred in fewer than 5% of the stomachs were not included in the IRI calculations. We calculated an overall index of fullness for each monthly sample by dividing the mean weight of stomach contents for that month by the mean fork length of all fish examined that contained food and multiplying this value by 100. We used one-tailed Mann-Whitney U tests ($\alpha = 0.05$) to compare mean indices of fullness between years.

To analyze dietary similarity among the three species, we calculated Morisita's index of overlap, described by Horn (1966):

$$C_x = \frac{2 \sum_{i=1}^s x_i y_i}{\sum_{i=1}^s x_i^2 + \sum_{i=1}^s y_i^2}$$

where

C_x = overlap coefficient,

x_i = proportion of the total diet of fish species x contributed by the food category i , and

y_i = proportion of the total diet of fish species y contributed by food category i .

A C_x value of 0 would indicate no food categories in common, while 1.0 would indicate identical diets. Zaret and Rand (1971) assumed a value ≥ 0.6 indicated significant overlap, which we also used for our comparisons.

RESULTS

We examined stomachs from 468 chinook salmon, 227 steelhead, and 337 Sacramento squawfish (Table 1). Adequate numbers of chinook salmon (minimum 14) were collected in all months with the exception of July 1992 and February 1993. Monthly mean lengths varied from 40 mm in February 1992 to 72 mm in June 1992. Target numbers of steelhead (14) were not collected until April in 1992 and May in 1993. Mean lengths ranged from 36 mm in April 1992 to 112 mm in July 1992. Squawfish numbers were adequate (12) in all months. On the basis of their unimodal length-frequency distribution, we judged nearly all the juvenile squawfish to be yearlings whose mean lengths ranged from 50 mm in February 1992 to 88 mm in June 1992.

Composition of Diets

Chinook salmon and steelhead utilized a variety of food items, but the major portion of their diet was composed of only a few taxa of aquatic insects. In 1992, major food items of chinook salmon were chironomid pupae, larvae, and adults; baetid adults; and hydropsychid larvae and pupae (Table 2). Their diet was less

Table 1. Monthly sample size and mean fork lengths (FL) of chinook salmon, steelhead, and Sacramento squawfish juveniles, lower American River, 1992 and 1993.

Species	Year		February	March	April	May	June	July
Chinook salmon	1992	Sample size	45	47	44	41	38	
		Mean FL	40	44	56	63	72	
		SD	8	10	13	8	9	
	1993	Sample size		53	66	64	49	21
		Mean FL		39	50	55	55	69
		SD		4	7	11	9	9
Steelhead	1992	Sample size			38	23	45	21
		Mean FL			36	89	82	112
		SD			2	7	8	10
	1993	Sample size				32	36	32
		Mean FL				43	56	61
		SD				8	10	15
Sacramento squawfish	1992	Sample size	18	31	48	27	30	45
		Mean FL	50	60	64	76	88	85
		SD	10	33	16	32	42	31
	1993	Sample size	12	20	46	22	22	16
		Mean FL	62	59	66	63	78	86
		SD	4	9	11	12	15	17

varied in 1993 when they fed primarily on chironomids. In general, chinook salmon fed on a greater diversity of items as the year progressed in 1992 (Fig. 3) but, in 1993, salmon fed primarily on chironomids in all months except February. Steelhead feeding followed a similar pattern in 1992, when baetid adults and nymphs and hydropsychid larvae were major food items; these items were uncommon in 1993 when chironomids were the major food every month (Fig. 4). In both years, terrestrial arthropods and fish larvae were uncommon in the diets of both chinook salmon and steelhead.

Sacramento squawfish fed mainly on aquatic insects in both years, primarily corixid adults and chironomid larvae; baetid adults and hydropsychid larvae also were prominent on the diet in 1992 (Table 2). Seasonally, chironomids decreased in importance from late winter through spring, while corixids increased in importance and by May had become a major food item (Fig. 3). Fish (Sacramento sucker larvae) were a common food item only in May 1993. Terrestrial arthropods (arachnids and formicids) were occasionally eaten, although these items formed a substantial part of the diet only in March 1993, when their collective IRI% was 44. Less common food items were lumbricids, decapods, and dytiscids.

Table 2. Major food items of juvenile chinook salmon, steelhead, and Sacramento squawfish in the lower American River, 1992-1993. Numbers represent percent index of relative importance (IRI%) for each year.

Species	Prey Item	Year	
		1992	1993
Chinook salmon	Baetid adults	20%	
	Hydropsychid larvae & pupae	12%	
	Chironomid larvae	11%	40%
	Chironomid pupae	50%	54%
	Chironomid adults		3%
	Other	7%	3%
Steelhead	Baetid nymphs	6%	
	Baetid adults	75%	
	Hydropsychid larvae	14%	
	Chironomid larvae		67%
	Chironomid pupae		29%
	Other	5%	4%
Sacramento squawfish	Baetid adults	7%	
	Hydropsychid larvae	5%	
	Corixid adults	66%	40%
	Chironomid larvae	13%	21%
	Chironomid adults		10%
	Fish larvae		23%
	Other	9%	7%

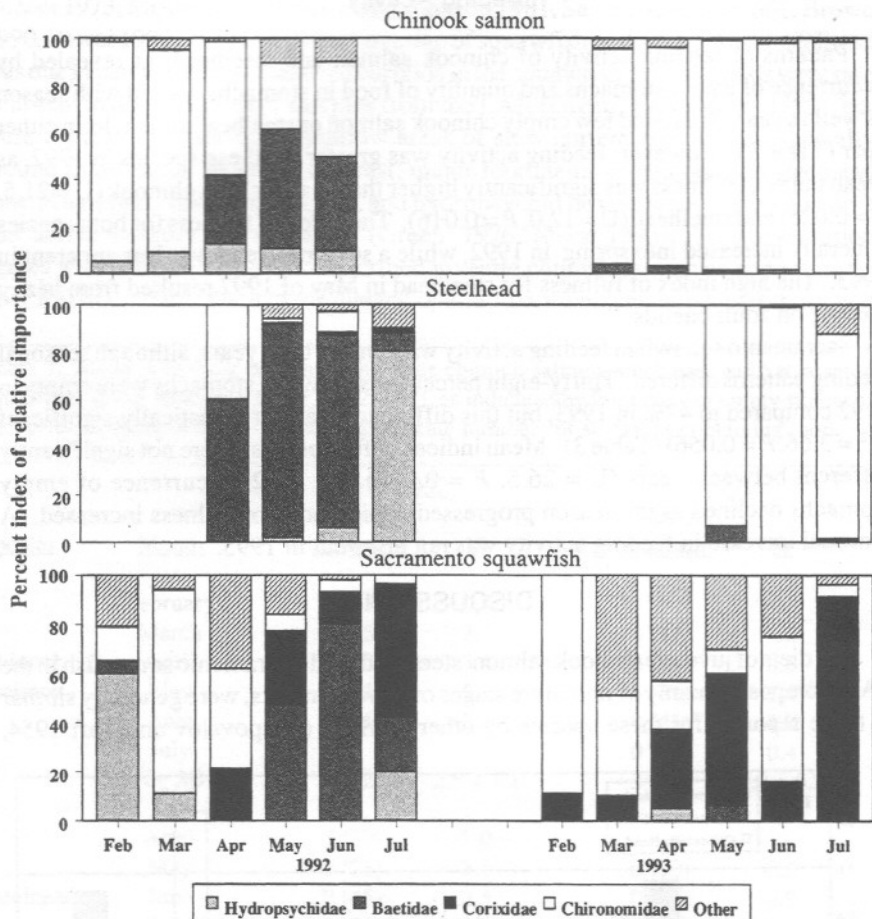


Figure 3. Major food items of chinook salmon, steelhead, and Sacramento squawfish in the lower American River by month, 1992 and 1993. Food items are represented by percent index of relative importance (IRI%) for each month.

Dietary Overlap

The diet of Sacramento squawfish did not significantly overlap ($C_x < 0.6$) with that of either chinook salmon or steelhead (Fig. 4). The greatest similarity between the squawfish and chinook salmon diet occurred in March 1992, when both species fed heavily on chironomids. The greatest similarity between squawfish and steelhead was in July 1992, when baetids were a common food item for both species.

Greater overlaps occurred between chinook and steelhead, especially in June 1992 and May and June 1993 (Fig. 4). In 1992, baetids constituted the major food item by weight for both species, while in 1993 chironomids (both larvae and pupae) formed the bulk of the diet for both species in these months.

Feeding Activity

Patterns of feeding activity of chinook salmon and steelhead, as revealed by occurrence of empty stomachs and quantity of food in stomachs, varied with season as well as year. We found few empty chinook salmon or steelhead stomachs in either year (Table 3). However, feeding activity was greater for these species in 1992, as mean index of fullness was significantly higher that year for both chinook ($U = 21.5$, $P = 0.028$) and steelhead ($U = 12.0$, $P = 0.016$). The index of fullness for both species generally increased into spring in 1992, while a seasonal trend was less apparent in 1993. The high index of fullness for steelhead in May of 1992 resulted from heavy feeding on adult baetids.

Sacramento squawfish feeding activity was similar both years, although seasonal feeding patterns differed. Thirty-eight percent of squawfish stomachs were empty in 1992 compared to 47% in 1993, but this difference was not statistically significant ($\chi^2 = 3.66$, $P = 0.056$) (Table 3). Mean indices of fullness also were not significantly different between years ($U = 26.5$, $P = 0.077$). In 1992, occurrence of empty stomachs declined as the season progressed, while index of fullness increased. A seasonal increase in feeding activity was not apparent in 1993.

DISCUSSION

The diets of juvenile chinook salmon, steelhead, and Sacramento squawfish in the LAR, composed mainly of immature stages of aquatic insects, were generally similar to those reported for these species by other workers (Shapovalov and Taft 1954,

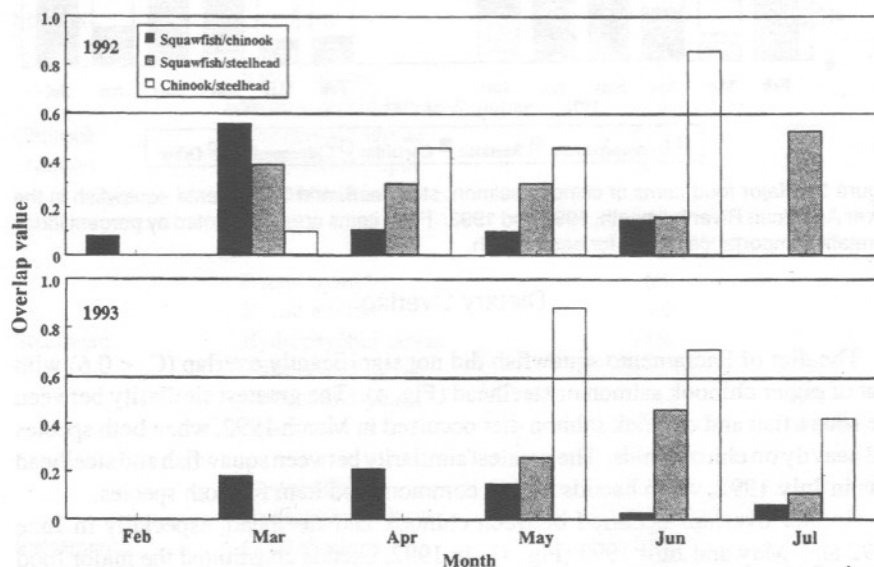


Figure 4. Morisita's overlap indices, by month, for diets of juvenile chinook salmon, steelhead, and Sacramento squawfish from the lower American River, 1992 and 1993. Solid horizontal line indicates level of significant overlap.

Becker 1973, Moyle 1976, Johnson and Johnson 1981, Sagar and Glova 1988, Brown 1990, Power 1992). The low dietary overlap of squawfish with chinook and steelhead was due in part to the importance of corixids in the squawfish diet. Corixids were rare or absent in drift and Surber samples from the LAR (Green⁵ 1976, Merz⁶ 1993), but were commonly observed in shallow areas of slow current near shore, especially around filamentous algae. In contrast, major food items of chinook and steelhead (baetid nymphs, chironomid larvae, and hydropsychid larvae) were common in drift and Surber samples in swift water (Green⁵ 1976, Merz⁶ 1993). This observation agrees with Moyle (1976) who described juvenile chinook salmon as opportunistic drift feeders. Thus, the lack of dietary overlap between squawfish and these two

Table 3. Monthly feeding patterns of juvenile chinook salmon, steelhead, and Sacramento squawfish, lower American River, 1992-1993, as indicated by frequency of empty stomachs and index of fullness (IF). Parentheses indicate numbers of stomachs containing food.

Species	Month	1992		1993	
		% Empty Stomachs	IF	% Empty Stomachs	IF
Chinook salmon	February	0 (45)	0.8		
	March	4 (45)	2.1	4 (51)	1.0
	April	2 (43)	3.4	0 (66)	0.8
	May	0 (41)	2.7	5 (61)	1.1
	June	5 (36)	3.3	8 (45)	1.0
	July			0 (21)	0.4
	$\bar{X} \pm SD$	2 ± 2	2.5 ± 1.1	3 ± 3	0.9 ± 0.3
Steelhead	April	3 (37)	1.0		
	May	0 (23)	7.1	6 (30)	0.6
	June	0 (45)	2.5	0 (36)	0.8
	July	0 (21)	3.7	3 (31)	0.9
	$\bar{X} \pm SD$	1 ± 2	3.6 ± 2.6	3 ± 3	0.8 ± 0.2
Sacramento squawfish	February	67 (6)	0.4	33 (8)	0.2
	March	61 (12)	0.2	55 (9)	0.2
	April	50 (24)	1.1	43 (26)	1.0
	May	19 (22)	0.7	64 (8)	0.2
	June	13 (26)	1.5	32 (15)	0.2
	July	20 (36)	1.4	56 (7)	1.4
	$\bar{X} \pm SD$	38 ± 24	0.9 ± 0.5	47 ± 14	0.5 ± 0.5

⁵ Green, D.B. 1976. Comparison of benthic macroinvertebrates above and below a storm drainage-sewage outfall in the American River, Sacramento, California. M.S. Thesis, California State University, Sacramento, California, USA.

⁶ Merz, J.E. 1993. A survey of drift and benthic communities and their use in the diet of the more abundant fish species in the lower American River, February-July, 1992. California State University, Sacramento, Foundation Contract Report FG1353.

salmonids is likely due to differences in their specific feeding locations. The dietary overlap between chinook salmon and steelhead was highest in May and June, the time when steelhead juveniles were most abundant and chinook salmon juveniles were decreasing in abundance due to downstream migration.

The contrasting environmental conditions of the LAR between 1992 and 1993, namely of flow variability and water temperature, resulted in different feeding patterns, at least for chinook salmon and steelhead. Highly fluctuating water levels and lower water temperatures in 1993 probably were responsible for the reduced feeding that year. Variable flows are generally detrimental to benthic invertebrates (Ward 1976, Petts 1984). Fluctuating discharges from a hydroelectric dam on the Connecticut River reduced macroinvertebrate populations in substrates that were repeatedly exposed (Fisher and LaVoy 1972); only chironomids were found in the zone that experienced the greatest exposure. Thus, it is not surprising that LAR chinook salmon, steelhead, and squawfish (particularly the former two species) greatly increased their utilization of chironomids in 1993.

In summary, our data support the hypothesis that fluctuating water levels during the salmonid nursery period inhibit feeding of chinook salmon, steelhead, and possibly Sacramento squawfish. Future studies are needed on responses of macroinvertebrate populations to levels of river discharge and degree of water fluctuations and how changes in dietary patterns of fishes affect growth and survival.

ACKNOWLEDGMENTS

We thank W. Snider, California Department of Fish and Game, for his support on this project and his LAR crew for assistance in field collections; W. Fields, Hydrozoology, Inc. and W. Shepard, California State University, Sacramento, for verification of insect identifications; and D. Faoro for use of his computer program on calculation of dietary overlap. We also thank L. Brown, U.S. Geological Survey; P. Moyle, University of California, Davis; D. Kohlhorst, California Department of Fish and Game; and two anonymous reviewers whose critical reviews greatly improved the manuscript.

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Received: 17 August 1995

Accepted: 22 June 1996